Estimated time needed: 10 minutes

In this lab, you will apply the concepts that you learnt in this module to a relational database schema called Car Dealership, which is designed to keep track of automobile sales in a car dealership.

Objectives:

After completing this lab, you will be able to:

- Evaluate your knowledge of Relational Database Concepts and the Entity-Relationship (ER) Diagram
- Imporve your understanding of terms related to relational models like entity, attribute, and keys.

::page{title=" Concepts covered in the lab" }

1. `Entity`: A noun: person, place, or thing

2. `Attributes`: The data elements that characterize the entity and tell us more about the entity.

3. `Primary key`: Uniquely identifies each tuple or row in a table and provides a way of defining relationships between tables

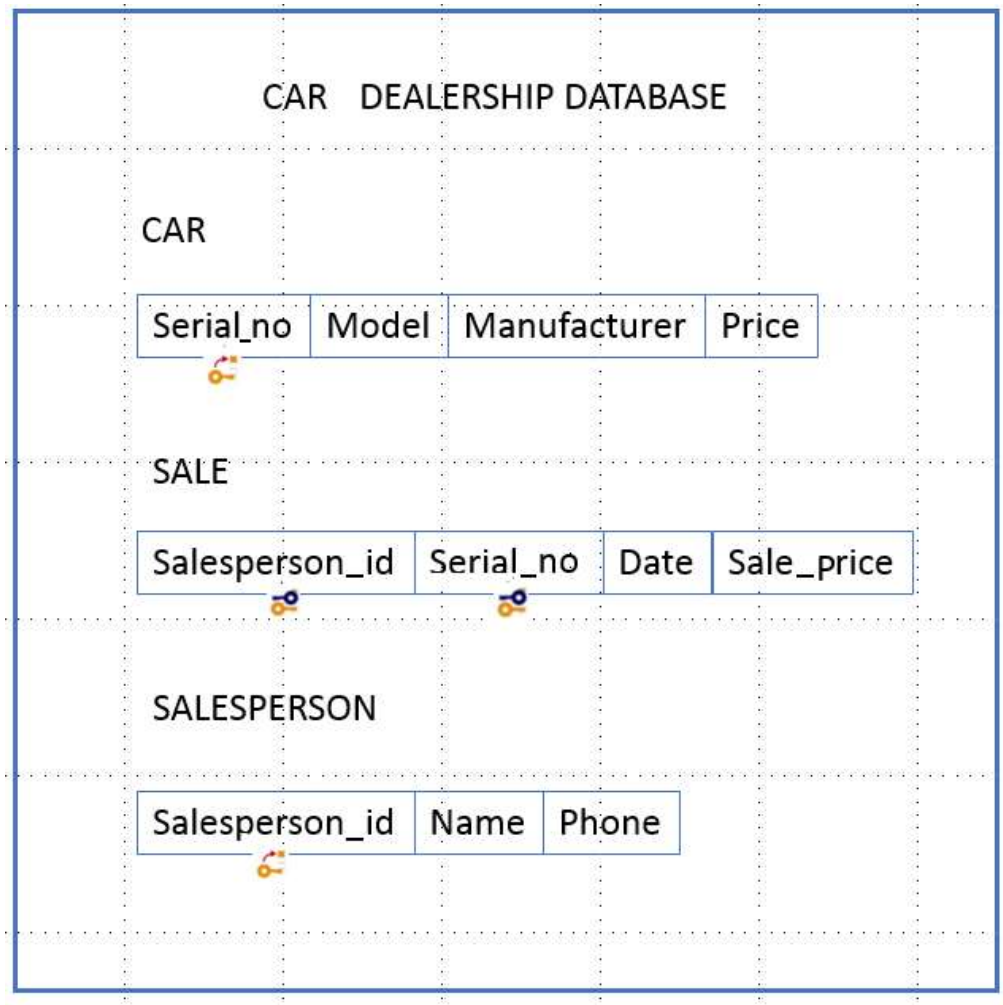
4. `Foreign key`: Primary keys defined in other tables, creating a link between the tables.

5. `Entity relationship (ER) diagram`: Represents entities called tables, and their relationships. The building blocks of an ER diagram are entities and attributes.

::page{title=" Exercise" }

In this exercise, we will be working on a relational database schema called Car Dealership. A database has to be designed to keep track of automobile sales in a car dealership.

Schema diagram for the Car Dealership relational database:



Relational instance of SALE:

Salesperson_id	Serial_no	Date	Sale_price
10001	1we4ds87	12/03/2020	$ 10,000.00
10005	d63jw3ty	12/03/2020	$ 5,000.00
10009	sy63bjd1	13/03/2020	$ 25,000.00
10001	k2k4edr8	13/03/2020	$ 49,000.00
10051	w3r334ac	13/03/2020	$ 8,000.00

Now let us go through some questions based on the above database schema of Car Dealership and relational instance of SALE:

1. How many relations does the Car Dealership database schema contain?

   - Hint
   - Answer

2. How many columns does the relation Car contain?

   - Hint
   - Answer

3. How many rows does the relation Sale contain?

   - Hint
   - Answer

4. Identify the attributes of the relation Salesperson.

   - Hint
   - Answer

5. Identify which relations of the Car Dealership database have primary keys. Name the primary keys if exist.

   - Hint
   - Answer

6. Identify which relations of the Car Dealership database have foreign keys. Name the foreign keys if exist.

   - Hint
   - Answer

::page{title=" Summary" }

Congratulations!

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Changelog

Date	Version	Changed by	Change Description
2023-07-12	5.0	K Sundararajan	Minor update based on SME feedback
2023-07-07	4.0	K Sundararajan	Updated lab content based on SME's feedback
2023-07-04	3.0	K Sundararajan	Small update for content rendering
2021-03-05	2.0	Sandip Saha Joy	Created revised md version
2018	1.0	Rav Ahuja	Created initial version

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